MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

GEARS

TO ALL HOLDERS OF MIL-STD-35-49A(MI):

1. THE FOLLOWING PAGES OF MIL-STD-35-49A(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
3	30 JUNE 1992	3	28 SEPTEMBER 1984
4	30 JUNE 1992	4	28 SEPTEMBER 1984
31	28 SEPTEMBER 1984	(Reprinted without change)	
32	30 JUNE 1992	32	28 SEPTEMBER 1984
33	30 JUNE 1992	33	28 SEPTEMBER 1984
34	30 JUNE 1992	34	28 SEPTEMBER 1984
35	30 JUNE 1992	35	28 SEPTEMBER 1984
36	28 SEPTEMBER 1984	(Reprinted without change)	
37	28 SEPTEMBER 1984	(Reprinted without change)	
38	30 JUNE 1992	38	28 SEPTEMBER 1984
39	30 JUNE 1992	39	28 SEPTEMBER 1984
40	28 SEPTEMBER 1984	(Reprinted without change)	

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of MIL-STD-35-49A(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Army-MI

Review activities:
Army-AT
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(Project EGDS-A421)

AREA EGDS

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating Cadmium (Electro-Deposited)

MILITARY

MIL-I-17214	Indicator, Permeability Low Mu-Go-No-Go

STANDARDS

FEDERAL

FED-STD-151	Metal, Test Methods

MILITARY

DOD-STD-35	Automated Engineering Document Preparation System
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods of Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1285	Marking of Electrical and Electronic Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AGMA201.02	Tooth Proportions for Coarse-Pitch Involute Spur Gears
AGMA203.03	Fine Pitch On-Center Face Gears for 20 Degree Involute Spur Pinions
AGMA207.06	Tooth Proportions for Fine-Pitch Involute Spur and Helical Gears
AGMA2005-B	Design Manual for Bevel Gears
AGMA374.04	Design Manual for Fine-Pitch Gearing
AGMA390.03	AGMA Gear Handbook-Volume 1-Gear Classification, Materials and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers Association, 1901 North Fort Myer Drive, Arlington, Va. 22209.)

NAS276	Gear-Spur, 20 Degree Involute, Low Inertia, 48 and Finer Pitch
NAS727	Gear-Spur, 20 Degree Involute, Low Inertia, 48 and Finer Pitch

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

ASTMB117	Standard Test Method of Salt Spray (Fog) Testing
ASTME380	Standard Practice for use of the International System of Units (SI) (The Modernized Metric System)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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CHAR CODE	STANDARD PARAGRAPH
1792	THE BACKLASH ALLOWANCE SHALL BE 1792 MICROINCH(S) PER GEAR.
1950	THE BACKLASH ALLOWANCE SHALL BE 1950 MILLIMETER(S) PER GEAR.
1794	THE BACKLASH TOLERANCE SHALL BE NOT GREATER THAN 1794 INCH(S).
1952	THE BACKLASH TOLERANCE SHALL BE NOT GREATER THAN 1952 MILLIMETER(S).
1796	THE BACKLASH TOLERANCE SHALL BE NOT LESS THAN 1796 INCH(S).

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
DESIGN AND CONSTRUCTION (CONT)	
MINIMUM (CONT)	
SI	
(MILLIMETER)	1954
GEAR TOOTH FORMS AND GOVERNING STANDARDS (SELECT ONE)	
FULL DEPTH INVOLUTE PER AGMA201.02 (GEAR NUMBER)	4201
COURSE PITCH INVOLUTE PER AGMA201.02, 20 DEGREES (GEAR NUMBER)	4203
COURSE PITCH INVOLUTE PER AGMA201.02, 25 DEGREES (GEAR NUMBER)	4204
SPIRAL BEVEL PER AGMA2005-B (GEAR NUMBER)	4207

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CHAR CODE	STANDARD PARAGRAPH
1954	THE BACKLASH TOLERANCE SHALL BE NOT LESS THAN 1954 MILLIMETER(S).
	GEAR TOOTH FORMS AND GOVERNING STANDARDS. THE GEAR TOOTH FORMS AND GOVERNING STANDARDS SHALL BE AS FOLLOWS:
4201	GEAR NUMBER(S) 4201, SHALL BE FULL DEPTH INVOLUTE, 14.5 DEGREES, IN ACCORDANCE WITH AGMA201.02.
4203	GEAR NUMBER(S) 4203, SHALL BE COARSE PITCH INVOLUTE, 20 DEGREES, IN ACCORDANCE WITH AGMA201.02.
4204	GEAR NUMBER(S) 4204, SHALL BE COARSE PITCH INVOLUTE, 25 DEGREES, IN ACCORDANCE WITH AGMA201.02.
4207	GEAR NUMBER(S) 4207, SHALL BE SPIRAL BEVEL, IN ACCORDANCE WITH AGMA2005-B.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
DESIGN AND CONSTRUCTION (CONT)	
GEAR TOOTH FORMS AND GOVERNING STANDARDS (CONT)	
FACE TYPE WITH SPUR PINION PER AGMA203.03 (GEAR NUMBER)	4209
STUB INVOLUTE PER AGMA201.02 (GEAR NUMBER)	4212
FINE PITCH INVOLUTE PER AGMA207.06 (GEAR NUMBER)	4214
STRAIGHT BEVEL PER AGMA2005-B (GEAR NUMBER)	4216
WORMGEAR PER AGMA374.04 (GEAR NUMBER)	4218
GEAR CONFIGURATION ___	1960
ABRASIVE RESISTANCE (GEARS ANODIZED AFTER FINAL MACHINING).	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ___	1868
NO BASE ALUMINUM VISIBLE.	1870

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CHAR CODE	STANDARD PARAGRAPH
4209	GEAR NUMBER(S) 4209, FACE TYPE, 20 DEGREES INVOLUTE WITH SPUR PINION IN ACCORDANCE WITH AGMA203.03.
4212	GEAR NUMBER(S) 4212, SHALL BE STUB INVOLUTE, 20 DEGREES, IN ACCORDANCE WITH AGMA201.02.
4214	GEAR NUMBER(S) 4214, SHALL BE FINE PITCH INVOLUTE, 20 DEGREES, IN ACCORDANCE WITH AGMA207.06.
4216	GEAR NUMBER(S) 4216, SHALL BE STRAIGHT BEVEL, IN ACCORDANCE WITH AGMA2005-B.
4218	GEAR NUMBER(S) 4218, SHALL BE WORMGEAR, IN ACCORDANCE WITH AGMA374.04.
1960	THE GEAR(S) CONFIGURATION SHALL BE 1960.
1868	ABRASIVE RESISTANCE (GEARS ANODIZED AFTER FINAL MACHINING). THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1868.
1870	ABRASIVE RESISTANCE (GEARS ANODIZED AFTER FINAL MACHINING). THERE SHALL BE NO VISIBLE BASE ALUMINUM METAL WHEN THE ANODIC COATING AS VIEWED UNDER A 25 POWER MICROSCOPE.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
ABRASIVE RESISTANCE (CONT)	
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ___	1862
TABOR ABRASOR DEVICE.	1864
CHEMICAL COMPOSITION	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ___	1962
CERTIFICATE OF COMPOSITION.	1964
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ___	1712
PER FED-STD-151, METHOD 111.2.	1714
PER FED-STD-151, METHOD 112.2.	1716
SUPPLIERS CERTIFICATION.	1718

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CHAR CODE	STANDARD PARAGRAPH
1862	ABRASIVE RESISTANT (GEARS ANODIZED AFTER FINAL MACHINING). TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1862.
1864	ABRASIVE RESISTANCE (GEARS ANODIZED AFTER FINAL MACHINING). THE TEST SPECIMEN SHALL BE CAPABLE OF RESISTING 1000 CYCLES OF A LOAD OF 250 GRAMS ON A NO. CS-10 CALIBRASE WHEEL WITHOUT DESTRUCTION OF THE ANODIC COATING. A TABOR ABRASOR IS CONSIDERED A SUITABLE DEVICE FOR PERFORMING THIS TEST.
1962	CHEMICAL COMPOSITION. REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1962.
1964	CHEMICAL COMPOSITION. SUPPLIER SHALL FURNISH A CERTIFICATE OF COMPOSITION.
1712	CHEMICAL COMPOSITION. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1712.
1714	CHEMICAL COMPOSITION. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH METHOD 11.2 OF FED-STD-151.
1716	CHEMICAL COMPOSITION. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH METHOD 112.2 OF FED-STD-151.
1718	CHEMICAL COMPOSITION. SUPPLIER°S CERTIFICATION MAY BE ACCEPTED IN LIEU OF CHEMICAL ANALYSIS TO VERIFY CONFORMANCE WITH MATERIAL SPECIFIED.

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CHARACTERISTIC NAME	CHAR CODE
EMBRITTLEMENT RELIEF	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1908
NO EMBRITTLEMENT.	1910
PER QQ-P-416, TREATMENT.	1912
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1902
PER QQ-P-416.	1904
FUNGUS	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4760
FUNGUS RESISTANT BY CERTIFICATION OR BY TEST.	4761

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CHAR CODE	STANDARD PARAGRAPH
1908	EMBRITTLEMENT RELIEF. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1908.
1910	EMBRITTLEMENT RELIEF. THE TEST SPECIMEN SHALL EXHIBIT NO EMBRITTLEMENT FROM THE PLATING PROCESS.
1912	EMBRITTLEMENT RELIEF. THE GEAR SHALL BE SUBJECTED TO THE HYDROGEN EMBRITTLEMENT RELIEF TREATMENT OF QQ-P-416.
1902	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1902.
1904	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH QQ-P-416.
4760	FUNGUS. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 4760.
4761	FUNGUS. ALL EXTERNAL MATERIALS SHALL BE NONNUTRIENT TO FUNGUS GROWTH OR SHALL BE SUITABLY TREATED TO RETARD FUNGUS FROWTH. THE MANUFACTURER SHALL VERIFY BY CERTIFICATION THAT ALL EXTERNAL MATERIALS ARE FUNGUS RESISTANT OR SHALL PERFORM THE FUNGUS TEST. WHEN THE FUNGUS TEST IS PERFORMED, THERE SHALL BE NO EVIDENCE OF FUNGUS GROWTH ON THE EXTERNAL SURFACES.

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CHARACTERISTIC NAME	CHAR CODE
FUNGUS (CONT)	
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4754
PER MIL-STD-810, METHOD 508.	4755
HUMIDITY	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4798
NO EVIDENCE OF CORROSION OR PHYSICAL DAMAGE.	4799
NO DEGRADATION OF PERFORMANCE.	4815
REMAIN WITHIN SPECIFIED TOLERANCE.	4791
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4788
PER MIL-STD-810, METHOD 507.	4871
*PROCUREMENT NUMBER (SELECT ONE)	
TEST PROCEDURE I.	4767

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